

SciGuard: response to the supplied review reports

Prepared September 27, 2026. This responds to the user's simulated multi-agent case review; it is not an actual editorial decision or a submitted response to JMLR.

Overall response

We agree that a major revision was warranted. The revision is not limited to prose or adding another notebook cell. It replaces the competing implementations with one shared codebase, reruns the available evidence, changes unsupported claims, and makes the missing external validation explicit. It does **not** claim that the outstanding JMLR-level novelty or realistic deployment-evidence concerns have been fully resolved.

The strongest new finding is unfavorable to a universal multiplier recommendation. In the 500-replication primary rerun its simultaneous coverage is 0.932; in the 200-replication rare-event stress design it is 0.775. Exact Bonferroni coverage is 0.960 and 0.995, respectively. These are simulation estimates with Monte Carlo uncertainty, not universal probabilities. The complete results and intervals appear in the manuscript and current CSVs.

1. Novelty and the original theorem

Concern: The previous theorem essentially assumed the approximation it needed; SAR was presented too strongly as a new statistical invention.

Action: The introduction, abstract and discussion now call SciGuard an applied-methods and reproducibility framework. The paper explicitly identifies exact binomial bounds, Hoeffding, Holm, multiplier inference, the Lipschitz extension, pilot splitting and alpha spending as established constructions. The finite-family exact guarantee has a direct proof. The multiplier statement has a fixed-dimensional CLT/covariance/quantile argument, with its positive-variance and bootstrap-quantile requirements spelled out.

The high-dimensional discussion retains a conservative sufficient regime and acknowledges later CCKK improvements. It does not claim an unproved sharp grouped-data rate. The finite environment problem is explicitly recognized as an LTT instantiation, not separated from LTT by a misleading distinction between “parameters” and “environments.”

Status: Reframing and technical clarification implemented. A new minimax theorem, sharp non-asymptotic multiplier error rate, or optimal adaptive/sequential method remains open.

Location: Manuscript Sections 1-4; notebook Section 2; `core.py`, `baselines.py`.

2. Shared transformations and dependence

Concern: Reusing observations across stresses was described only as correlation.

Action: A row-construction lemma now writes each entire loss vector as a fixed measurable function of one independent source row and its stress randomness. This proves row independence while allowing arbitrary within-row dependence. Deterministic dependence between coordinates does not itself invalidate a vector CLT. Conversely, cross-row normalization, repeated patients, clusters and temporal dependence are explicitly outside that simple argument.

One multiplier is shared across all coordinates in a row. Grouped losses use ratio-estimator influence vectors on a master sample. Unrelated environment samples are not artificially paired or truncated. Tests include duplicated columns, a single subgroup versus the paired method, unequal sample sizes, and covariance-versus-row Gaussian sampling.

Status: Implemented for the stated sampling assumptions. Actual EHR row independence and temporal/hospital clustering are not established without the real-data audit.

Location: Manuscript Section 3 and Proposition 3; notebook Sections 2.1 and 11.1; `core.py`, `groups.py`.

3. Finite-sample guarantees and unstable rare-event behavior

Concern: Asymptotic coverage was being used as a broad practical assurance claim.

Action: Added exact one-sided Clopper-Pearson Bonferroni bands, Hoeffding bands for bounded loss, and exact-binomial Holm tests. Normal Bonferroni is named explicitly so it cannot be confused with exact inference. The multiplier no longer raises an exception when all columns are constant; it uses conservative bounds. The grouped method no longer automatically certifies zero-error groups using a zero standard error.

Added continuous bounded losses, rare binary errors, and a 125-environment stress family. The observed multiplier undercoverage is retained, explained, and reflected in the abstract and method guidance. No empirical-variance floor is presented as a full correction for low-event-count failures. The procedure must be selected prospectively; taking the most favorable bound afterward is not an allowed shortcut.

Status: Implemented. Finite-sample claims still depend on the stated independent-sampling and bounded/Bernoulli-loss model. Unbounded heavy tails and adversarial adaptive shifts are not covered by these experiments.

Location: Manuscript Sections 4 and 6.4; notebook Sections 2 and 8; `core.py`; `tests`.

4. Empirical centerpiece and training variability

Concern: A hand-constructed risk surface dominated the paper, while fitted-pipeline evidence used few refits.

Action: The fitted-model study now precedes calibration in the manuscript and uses 200 independent refits rather than 80. It uses shared training/assurance/reference rows across logistic regression and histogram gradient boosting, while keeping the different sample roles independent. The exact inference alternatives receive identical assurance losses. Model-fitting time and nominal reference risk are exported. The copula experiment is labeled calibration-only.

Training variability is represented by repeated fitted pipelines, not claimed to be a new unconditional finite-sample result for the multiplier. The revised protected feature and shared stress coupling change the DGP; old and new numbers are not treated as identical designs.

Status: Implemented for controlled simulations. This does not replace natural-shift evidence.

Location: Manuscript Section 6.1; notebook Section 4; `experiments.py`, `simulation.py`.

5. “True” risks estimated with too few reference rows

Concern: The earlier 1,500-row Monte Carlo estimate was effectively used as exact truth.

Action: Increased the independent reference sample to 20,000 rows and added simultaneous two-sided exact binomial intervals. The reference error budget is 0.01 over both models in each nested refit, and 0.01 over all four mechanisms in each mechanism refit. Exported environment-level intervals distinguish confirmed-safe, confirmed-unsafe and ambiguous cells.

Point-reference containment/ARR/UCP are labeled accordingly. Lower and upper containment indicators bound the unknown indicator on the reference-coverage event; their mean bracket is shown separately from Wilson intervals over replications. The paper does not claim that the reference brackets are confidence intervals for true coverage or simultaneously valid over all refits. All model/method settings also retain certified counts.

Status: Implemented uncertainty accounting. More reference sampling can still be useful near the tolerance; uncertainty has not been declared eliminated.

Location: Manuscript equation (reference-containment construction) and Tables 1/3 as numbered by the final compile; notebook Section 4; `metrics.py`.

6. MAR, MNAR and structured missingness

Concern: Mechanisms were arbitrarily parameterized, and the original MAR code could mask the variable on which missingness depended.

Action: Feature 0 is always observed across mechanism comparisons. MAR depends only on that driver. MNAR depends on the feature being hidden. Intercepts are calibrated against the known simulation distribution with fixed Gaussian quadrature rather than each evaluation batch. Tests verify that masking a subset of rows produces exactly the same masks as those rows in a full batch, eliminating the former cross-row normalization.

All mechanisms use severity as marginal missingness among the nine eligible features. Structured block failure maintains those marginals. The study now has 100 refits, exports realized rates, and compares multiplier, exact Bonferroni and exact Holm. MNAR results remain sensitivity scenarios and do not identify any actual clinical missingness law.

Status: Implemented. General conclusions about which mechanism is hardest are not claimed.

Location: Manuscript Section 6.2; notebook Section 5; `simulation.py`.

7. ATC comparison fairness

Concern: Reporting ATC unsafe-certification rates alongside formally controlled procedures invited an apples-to-oranges superiority claim.

Action: Removed ATC from the certification comparison. It now appears in a separate table of prediction MAE, bias and RMSE, using source-only calibration and target confidence scores. The implementation handles tied confidence values consistently with a strict threshold. Exact Holm and exact Bonferroni provide same-target certification comparators.

Status: Implemented. No invented “confidence-corrected ATC” guarantee was added.

Location: Manuscript Section 6.1 ATC paragraph/table; notebook Section 4.1; `baselines.py`.

8. When is multiplier computation worthwhile?

Concern: The original ARR improvement over Bonferroni was modest and insufficiently explained.

Action: Added paired ARR-gain estimates with MCSE against normal Bonferroni, exact Bonferroni and exact Holm across n /correlation settings. At $n=1,000$, the paired gain over normal Bonferroni increases from approximately 0.0005 at latent correlation 0.10 to 0.066 at 0.90. This is a finding within the chosen simulation, not a universal cutoff.

The practical rule now starts with required validity: prefer exact binary-loss methods when finite-sample control is required; consider multiplier efficiency only where its approximation is adequately supported. No near-nominal assertion is carried across rare-event regimes. The HGB nested example also shows exact Holm can nearly match multiplier recall.

Status: Implemented quantitative comparison and scoped guidance. No minimax-optimality or universal break-even correlation is claimed.

Location: Manuscript Section 6.4 and additional paired-gain table; notebook Section 7.

9. Environment design and Lipschitz uncertainty

Concern: Axes/ranges/resolution were left to users, and an unknown smoothness constant was effectively assumed.

Action: Added a concrete independent-pilot selection rule, a full simulation of the selected finite family, a conservative sample-size planning table, and explicit instructions about prespecification and independent assurance. The planning inequality controls bound width, not recall. It does not choose scientifically relevant axes without external knowledge.

The continuous experiment retains an analytically valid constant for its known bilinear surface. Half/twice-constant outputs are labeled sensitivity analyses. The text explicitly rejects estimating a global upper Lipschitz constant from maximum finite-grid slopes, even with sample splitting. It gives the α -plus- δ error allocation if a separately justified random global bound is available.

Status: Implemented practical workflow and corrected interpretation. A general globally valid Lipschitz estimator or optimal adaptive design remains open.

Location: Manuscript Section 5; notebook Sections 9-10; `design.py`, `continuous.py`.

10. Repeated and sequential certification

Concern: Static assurance does not address repeated model updates.

Action: Added the standard alpha-spending construction $\alpha_t = \alpha/[t(t+1)]$ with fresh assurance data and conditional finite-sample inference. Its summable budget supplies a simple valid route; it is not presented as a new efficient sequential theory or an evaluated dynamic deployment system.

Status: Basic conservative argument implemented; advanced sequential SAR development remains outside the current empirical study.

Location: Manuscript Section 4.3; notebook Section 10; design.py.

11. Model breadth, computation and ablations

Concern: Few model classes, insufficient bootstrap/grid/sample-size ablations, no timing.

Action: WDBC now includes logistic regression, Extra Trees and histogram gradient boosting. The package includes n /dependence sensitivity, equally spaced grid-size ablations, repeated bootstrap sizes/seeds and runtime summaries. The primary nonuniform 36-cell grid is explicitly distinguished from the equally spaced 6-by-6 grid. The Gaussian covariance shortcut and its computational complexity are documented and tested against row multipliers.

Status: Implemented for tabular models. No deep-network or large-scale performance claim is made. The 125-cell stress test enlarges K ; it is not a high-dimensional continuous geometry study.

Location: Manuscript Sections 6-7 and computational appendix; notebook Sections 6-8.

12. A genuine external distribution-shift benchmark

Concern: WDBC is small and semisynthetic; a real natural-shift benchmark is missing.

Action: Repaired and consolidated the MIMIC-IV protocol. Each admission is aligned to the patient's anchor year before defining approximate temporal intervals. Ambiguous source/deployment overlap is excluded. The first eligible adult admission is selected before temporal filtering; top-coded age is retained. Outcome/length-of-stay variables are excluded from predictors. Native and synthetic missingness are separately reported.

One family covers all models/domains/missingness. The protocol adds exact alternatives, small-group handling, cohort flow, an always-survive baseline, prevalence and imbalance diagnostics, a SAR map, and frozen-config/source-file hashes. The arbitrary 0.35 default has been removed; a prospective clinical risk tolerance and rationale must be supplied locally.

Status: OPEN EMPIRICAL REQUIREMENT. Credentialed data were not attached, so no real MIMIC result has been generated. Synthetic fixtures test software, not benchmark validity. Admission-table features remain a limited baseline; clinical timing, temporal support, dependence, representativeness and utility require an actual-data audit. This major-review request cannot honestly be marked resolved yet.

Location: Manuscript Section 8; notebook Section 11; mimic.py, config/mimic_protocol.json.

13. Related work and safe-operating-envelope precedents

Concern: Nearby literatures and safety-envelope work were missing or only name-checked.

Action: Added weighted conformal, PAC-Bayes and non-CCK simultaneous-band context. The discussion now compares the target and mechanism with perception-plus-planner safety, operational-domain monitoring, and causal online safe-region learning. LTT is described as a direct framework for the finite environment problem. Sources were checked against primary papers/documentation and recorded in SOURCE_NOTES.md.

Status: Implemented focused positioning; an exhaustive priority claim is not made.

Location: Manuscript Section 2; bibliography; SOURCE_NOTES.md.

14. Writing and reproducibility

Concern: Repeated explanations, duplicate/irregular headings, weak figure intuition, and inconsistent notebook/package/manuscript results.

Action: Replaced the manuscript with one coherent current source, a shorter contribution statement, explicit limitations, and current CSV-derived tables/macros. Added a workflow schematic, consistent method labels,

Wilson intervals and reference-uncertainty captions. The original versions remain in a historical archive. The canonical notebook now imports the package and has sequential sections, no shadowed statistical functions, no premature “analysis complete” declaration, and no silent fallback to unexecuted archived results.

Full/smoke outputs are isolated. The notebook checks all sections and exports a provenance manifest. The test suite targets scientific failure modes rather than only shape checks. Only the current manuscript and canonical notebook should be used for the next review.

Status: Implemented; execution and artifact checks are recorded separately in `VALIDATION_REPORT.md`. Actual MIMIC execution remains pending.

Before another journal submission

Complete the natural-shift benchmark and examine its practical usefulness against the trivial baseline; audit clinical data timing and dependence; obtain a defensible risk requirement; decide with the supervisor whether the applied contribution is sufficient for the intended venue or whether substantial new theory is needed; establish a public code archive and verify author, funding and disclosure details. This revision does not promise acceptance or declare the paper ready for JMLR submission.

Canonical notebook consistency audit

Source audited: SciGuard_JMLR_Reproducible_Analysis_MIMIC_IV.ipynb from the uploaded ZIP. Its original bytes are retained under archive/pre_revision/.

Material findings in the original

1. Multiple top-level notebooks and two main manuscript sources competed for authority.
2. Core functions were copied into the notebook and differed from the package in variance normalization, input validation, degenerate-column behavior and output attribute names.
3. Later cells redefined `shift_X`, `missing_mask` and other helpers with different signatures.
4. The original MIMIC execution cell contained a saved `FileNotFoundError`; later package/status cells were unexecuted. The ZIP contained no completed MIMIC result CSV.
5. Late simulations silently loaded historical outputs unless an extension flag was set.
6. Section numbering skipped 5, duplicated 9, and declared analysis complete before later analyses.
7. The MAR driver was potentially masked, and per-batch probability rescaling linked rows.
8. Nested reference risks from 1,500 observations were used as if their threshold classifications were exact; no reference-risk confidence intervals were supplied.
9. ATC appeared alongside certification methods despite having no matching familywise guarantee.
10. The grouped package routine could certify a zero-error group with a zero uncertainty bound.
11. MIMIC grouping used the patient anchor period without admission-year alignment; low-count domains could disappear from the report; its illustrative tolerance lacked clinical justification.
12. Historical figures and repeated manuscript figures did not always have identical bytes.
13. The notebook and manuscript carried independent numerical targets and archived assertions, making it possible for figures, tables and computation to drift apart.

Original-to-current coverage map

Original content	Current canonical location	Treatment
Imports, paths and FAST_TEST	Section 1	Explicit full/smoke configuration; portable root discovery
Core inference and repeated helper definitions	Section 2 / shared package	Unified implementation; finite-sample methods added
True risk surface and copula calibration	Section 3	Recomputed; original data retained
ARR violin, simultaneous coverage and intervals	Section 3	Rebuilt from current replication rows
Nested refits and model variability	Section 4	200 refits, shared model comparisons, reference intervals
ATC	Section 4.1	Separate risk-prediction evaluation
Alternative missingness mechanisms	Section 5	Corrected MAR/row locality; 100 refits and realized rates
WDBC single-model and model comparisons	Section 6	Consolidated three-model study; exact baselines

Original content	Current canonical location	Treatment
Grid-resolution study	Section 6	Nonuniform primary versus uniform grids distinguished
Sample-size and correlation sensitivity	Section 7	Six methods; paired efficiency-gain estimates
Bootstrap sampler and complexity	Sections 2.1 and 8	Shared implementation plus timing/seed/B study
Continuous SAR	Section 9	Analytic bound, full rerun, figure and sensitivity exports
Practitioner environment-design guidance	Section 10	Independent-pilot experiment and sample-size planner
MIMIC loading and execution	Section 11	Admission-aligned local protocol; explicit pending status
Package check and reproducibility checks	Sections 1 and 12	Imports must succeed; end-of-run structural/hash checks
Session and output information	Section 12	Manifest with configuration, versions, times and hashes
Old external benchmark variants	Historical archive	Preserved; not falsely combined with current MIMIC evidence

What “complete” means here

The canonical notebook covers every study previously in the MIMIC-IV notebook and the review-driven additions listed above, using one package implementation and traceable current outputs. It is complete as an executable public/synthetic workflow plus a credentialed-data protocol. It is **not** a completed real-MIMIC empirical study. Keeping that distinction is part of consistency, not an omission hidden by loading a placeholder table.

The ZIP, not a notebook file alone, is the runnable deliverable: the notebook requires the bundled source modules, data and export scripts. Standalone notebook viewing is supported.

Revision validation report

Scope and outcome

The public-data and synthetic analysis is executed end to end. The canonical notebook has 30 cells, including 13 executable cells, and eight embedded figures. Its source imports the shared package used by the command-line analyses. MIMIC-IV is deliberately marked NOT RUN; no credentialed data were supplied and no clinical benchmark result is represented as real.

Scientific checks

The 18-test scientific-contract suite passes. It checks exact-binomial coverage and Holm familywise error by count enumeration, rare/zero-event behavior, shared Gaussian multipliers, unequal independent samples, overlapping groups, confidence ties, row-local missingness, reference-interval boundaries, continuous envelopes, alpha spending and the MIMIC loader. The MIMIC end-to-end fixture is synthetic software validation only; it is not clinical evidence.

The full configuration uses 500 calibration replications, 200 independently fitted nested replications, 100 mechanism replications, 300 replications per sensitivity setting and 200 continuous-envelope replications. Independent reference samples have 20,000 rows. Monte Carlo uncertainty and uncertainty in reference-risk classification are retained.

The main multiplier coverage estimate is 0.932 (nominal target 0.95); the rare-event stress estimate is 0.775. Exact Bonferroni estimates are 0.960 and 0.995 respectively. These findings are reported consistently, including in the abstract, rather than hidden by favorable examples.

Execution and reproducibility

Both the full command-line workflow and the canonical notebook were run. Notebook execution uses a fresh Python process with an in-process IPython shell, executing cells in order and capturing their actual text and image output. Kernel socket binding is unavailable in this runtime, so standard Jupyter-kernel execution could not be validated here. A standard nbclient backend is included for environments that permit kernel sockets. No outputs were fabricated.

The final notebook metadata identifies its execution backend, completion state and duration. The manifest records configuration, package versions, section timing, source hashes and result-file hashes. Tables and numeric manuscript macros are generated from current CSVs; table_provenance.json records their input hashes. The manuscript is compiled from LaTeX. Scientific figures and document pages are rendered and visually checked.

The ZIP contains the source, notebook, tests, data, results, figures, manuscript, response, protocol and historical archive. Use the ZIP for execution; the standalone notebook requires its bundled modules and data. Requirements record the tested environment, not a guarantee that every operating system or future dependency version behaves identically.

Remaining requirements

Run the credentialed MIMIC-IV protocol after prospectively choosing a defensible risk tolerance and rationale. Audit feature timing, temporal alignment, repeated-patient handling, dependence, cohort support and practical value against the trivial baseline. The approximate calendar interval padding is an explicit analysis choice requiring sensitivity analysis on actual data. The public/synthetic studies do not establish clinical deployment suitability.

The revision clarifies established theory and adds useful evidence; it does not supply a new minimax result or guarantee that the contribution meets JMLR's novelty standard. Author, affiliation, funding, disclosure and public-code-release details still require author verification.